

Short Answer Test
CSA1613 → (01-AT-1)

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1.) Hospital Data warehouse

(a) Snowflake Schema

	Visit_Fact	Doctor_Demo
Patient_join		
Patient_keys	PK Visit_keys	Doctor_keys
Patient_ID	FK Patient_keys	Doctor-ID
Patient_name	FK Doctor_keys	Doctor_name
Age		

b.) OLAP Operations

Starling cube

[Patient, Doctor, Department, Time]

- * Rollup Patient → All
- * Rollup Doctor → All
- * Rollup Time → year
- * Keep @ Department - level unchanged.
- * Aggregate using Average (Avg-charge).

Result:

[All, All, Department, year]

(1) Number of cuboids

Formula:

no of cuboids = n (levels of each dimension)

There are 4 dimensions.

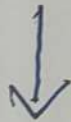
Each has 4 levels.

$$4 \times 4 \times 4 \times 4$$

$$= 256 \text{ cuboids}$$

2) Smart City Star Schema

Time_Dim



Traffic_Dim $\rightarrow$ Smart City Fact ---

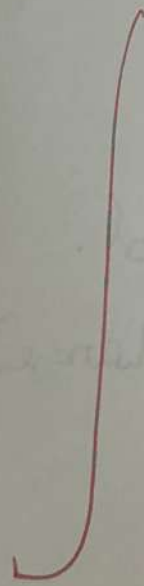
Transport Dim



Location_Dim

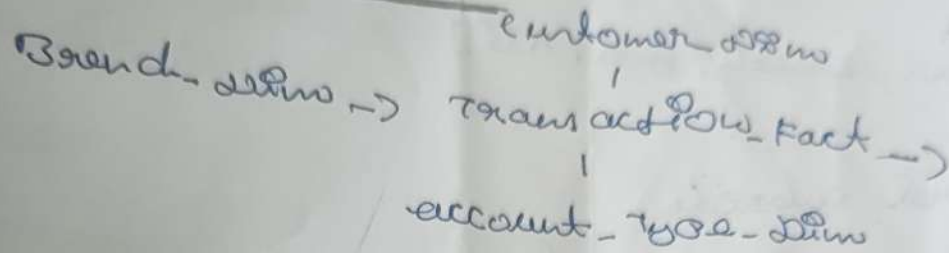


CellTower_Dim



1) Banking Data warehouse

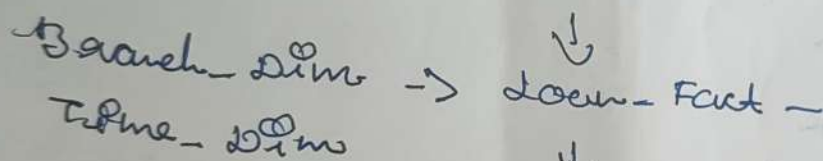
(a) Data Schema



measures:

transaction count
amount

customer-dim



Account_Type-dim

measures:

- Loan Amount
- Interest

2) OLAP Operation

[customer, Branch, Time, Account_Type]

operations:

* Roll-up customer → All

* Roll-up Account_Type → All

- * Roll up Time $\rightarrow$ year
- * Keep Branch

Result
[All, Branch, year, All]

(c) Total cuboids

Each dimension has 4 levels

$$4 \times 4 \times 4 \times 4 \\ = 256 \text{ cuboids}$$

4. Manufacturing companies

1. Roll up

summarize:

- * Select Price
- * Product $\rightarrow$ category
- * Time $\rightarrow$ current year
- * Factory $\rightarrow$ All

Result:

[category, year, All, factories]

2. Drill-Down

- * Department
- * Product type

down
* Electronic categories

* Pacific Region

Analysed:

Categories $\rightarrow$ Subcategories

SKU

Measure :-

* Sales Revenue

* Units sold.

3.) Slice

Condition:

* Gender = male

* Membership = Bronze

Display:

* Sales Revenue

Across:

* All stores.

2.1.200

- * specific data
- * selected product categories
- * specific time periods

Analysis:

- * sales revenue
- * units sold

